Light: Reflection and Refraction

Comprehensive Quick-Revision Guide covering Laws, Ray Diagrams, Formulations & Key Rules

1. Introduction to Light & Reflection

Light is a form of energy that enables us to see objects. It exhibits dual nature, but for ray optics, it is considered to travel in straight lines (rectilinear propagation).

Laws of Reflection

1. **First Law:** The angle of incidence (i) is always equal to the angle of reflection (r). That is, $\angle i = \angle r$.
2. **Second Law:** The incident ray, the reflected ray, and the normal to the reflecting surface at the point of incidence, all lie in the same plane.

Types of Images:

- **Real Image:** Formed when light rays actually intersect. Can be obtained on a screen. Always inverted.
- **Virtual Image:** Formed when light rays appear to intersect when produced backwards. Cannot be obtained on a screen. Always erect.

2. Spherical Mirrors

Spherical mirrors are curved reflecting surfaces which form part of a hollow sphere of glass.

- **Concave Mirror:** Reflecting surface is curved *inwards* (towards center of sphere). Converging mirror.
- **Convex Mirror:** Reflecting surface is curved *outwards*. Diverging mirror.

Key Terms Related to Spherical Mirrors

- **Pole (P):** Center of the reflecting surface of the mirror.
- **Center of Curvature (C):** Center of the hollow sphere of which mirror forms a part.
- **Radius of Curvature (R):** Radius of the sphere. Distance $PC = R$.
- **Principal Axis:** Straight line passing through Pole (P) and Center of Curvature (C).
- **Principal Focus (F):** Point on principal axis where parallel rays intersect (concave) or appear to diverge from (convex) after reflection.
- **Focal Length (f):** Distance between Pole (P) and Focus (F).

Relationship between Radius of Curvature & Focal Length: $R = 2f \quad \Rightarrow \quad f = R / 2$

3. Image Formation by Spherical Mirrors

Concave Mirror Summary Table

Position of Object	Position of Image	Size of Image	Nature of Image
At Infinity	At Focus (F)	Highly diminished, point-sized	Real and Inverted
Beyond C	Between F and C	Diminished	Real and Inverted
At C	At C	Same size as object	Real and Inverted

Position of Object	Position of Image	Size of Image	Nature of Image
Between C and F	Beyond C	Enlarged	Real and Inverted
At Focus (F)	At Infinity	Highly enlarged	Real and Inverted
Between P and F	Behind the mirror	Enlarged	**Virtual and Erect**

Convex Mirror Summary Table

Position of Object	Position of Image	Size of Image	Nature of Image
At Infinity	At Focus (F) behind mirror	Highly diminished, point-sized	Virtual and Erect
Between Infinity & Pole (P)	Between P and F behind mirror	Diminished	Virtual and Erect

USES **Concave Mirrors**

- Torches, searchlights, headlights (to get powerful parallel beams)
- Shaving mirrors (to see larger image)
- Dentists (to view large images of teeth)
- Solar furnaces (to concentrate sunlight)

USES **Convex Mirrors**

- Rear-view mirrors in vehicles (provides wider field of view and always erect image)
- Security mirrors in shops and blind corners of roads

4. Mirror Formula, Sign Convention & Magnification

New Cartesian Sign Convention (Mirrors & Lenses)

- Object is always placed to the **left** of the mirror/lens.
- All distances are measured from the **Pole (P)** / **Optical Center (O)**.
- Distances along incident light (to the right of origin) are **positive (+)**.
- Distances opposite to incident light (to the left of origin) are **negative (–)**.
- Distances perpendicular and above principal axis are **positive (+)**.
- Distances perpendicular and below principal axis are **negative (–)**.

Mirror Formula

$$1 / f = 1 / v + 1 / u$$

Where: u = object distance, v = image distance, f = focal length

Magnification (m) for Mirrors

$$m = h' / h = -\,(v / u)$$

Where h' = image height, h = object height.

- If m is **negative**, image is **Real & Inverted**.
- If m is **positive**, image is **Virtual & Erect**.
- If $|m| > 1$ (enlarged), $|m| = 1$ (same size), $|m| < 1$ (diminished).

5. Refraction of Light

Refraction is the bending of a light ray when it travels obliquely from one transparent medium to another due to a change in its speed.

- When traveling from **Rarer to Denser medium**: Light bends **towards the normal** ($\angle i > \angle r$).
- When traveling from **Denser to Rarer medium**: Light bends **away from the normal** ($\angle i < \angle r$).

Laws of Refraction (Snell's Law)

1. The incident ray, refracted ray, and normal at the point of incidence all lie in the same plane.
2. **Snell's Law:** The ratio of sine of angle of incidence to sine of angle of refraction is constant for a given pair of media and given wavelength.

$$\sin i / \sin r = \text{constant} = n_{21}$$

Refractive Index (n)

Refractive index represents the light-propagating ability of a medium.

- **Relative Refractive Index:** $n_{21} = v_1 / v_2$ (speed of light in medium 1 divided by speed in medium 2).
- **Absolute Refractive Index:** $n = c / v$, where c is speed of light in vacuum (3×10^8 m/s) and v is speed in medium.

6. Spherical Lenses

A transparent optical medium bounded by two spherical surfaces.

- **Convex Lens:** Thicker at middle, thinner at edges. *Converging lens*. Focal length f is **positive (+)**.
- **Concave Lens:** Thinner at middle, thicker at edges. *Diverging lens*. Focal length f is **negative (–)**.

Image Formation by Lenses

Convex Lens Summary Table

Position of Object	Position of Image	Size of Image	Nature of Image
At Infinity	At Focus F_2	Highly diminished	Real and Inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and Inverted
At $2F_1$	At $2F_2$	Same size	Real and Inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real and Inverted
At Focus F_1	At Infinity	Infinitely large	Real and Inverted
Between Focus F_1 and Optical Center O	On same side as object	Enlarged	**Virtual and Erect**

Concave Lens Summary Table

Position of Object	Position of Image	Size of Image	Nature of Image
At Infinity	At Focus F_1	Highly diminished	Virtual and Erect
Between Infinity and Optical Center O	Between F_1 and O	Diminished	Virtual and Erect

7. Lens Formula, Magnification & Power of Lens

Lens Formula

$$1/f = 1/v - 1/u$$

Magnification (m) for Lenses

$$m = h'/h = v/u$$

Power of a Lens (P)

Power is the degree of convergence or divergence of light rays achieved by a lens. It is defined as the reciprocal of focal length in meters.

$$P = 1/f \text{ (in meters)}$$

- **SI Unit:** Dioptre (D). $1\ D = 1\ m^{-1}$.
- **Convex Lens:** f is positive $\Rightarrow$ Power P is **Positive (+)**.
- **Concave Lens:** f is negative $\Rightarrow$ Power P is **Negative (–)**.
- **Combination of Lenses:** Net Power $P = P_1 + P_2 + P_3 + \ldots$